Hacker Rank (List of Assessments)

https://www.hackerrank.com/skills-verification

S.No.	Certification Name	Duration	No. of Questions	Assessment Topic
1	Python (Basic) Skills Certification Test	1 hr 30 mins assessment	Solve 2 questions	Covers topics like Objects and Types
2	Frontend Developer (React)	1 hr assessment	Solve 4 questions	Demonstrate your React, CSS, Javascript Skills
3	Software Engineer	1 hr 15 mins assessment	Solve 3 questions	Demonstrate your Problem Solving, SQL, REST API Skills
4	Software Engineer Intern	1 hr assessment	Solve 4 questions	Demonstrate your Problem Solving, SQL Skills
5	C# (Basic)	1 hr assessment	Solve 2 questions	Covers topics like Exception Handling, C# Program Structure, Properties and Indexers, Types and Variables
6	CSS (Basic)	20 mins assessment	Solve 8 questions	Covers topics like Cascading and Inheritance, CSS Layouts, Text Styling Fundamentals, Boxing of Elements in CSS,
7	Java (Basic)	1 hr assessment	Solve 6 questions	Covers topics like Classes, Inheritance, Data Structures, Exception Handling
8	JavaScript (Intermediate)	1 hr 30 mins assessment	Solve 2 questions	It covers topics like Design Patterns, Memory management, concurrency model, and event loops, among others.
9	JavaScript (Basic)	1 hr 30 mins assessment	Solve 2 questions	It covers topics like, Functions, Currying, Hoisting, Scope, Inheritance, Events and Error Handling.
10	Node (Basic)	1 hr assessment	Solve 2 questions	It covers topics like Package and Modules Management, Callbacks, Event Loop, Event Emitter, Buffers, Streams and File Systems
11	Problem Solving (Basic)	1 hr 30 mins assessment	Solve 2 questions	It covers basic topics of Data Structures (such as Arrays, Strings) and Algorithms (such as Sorting and Searching).
12	Problem Solving (Intermediate)	1 hr 30 mins assessment	Solve 2 questions	It covers topics of Data Structures (such as HashMaps, Stacks and Queues) and Algorithms (such as Optimal Solutions).

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S.No.	Certification Name	Duration	No. of Questions	Assessment Topic
13	React (Basic)	1 hr 30 mins assessment	Solve 2 questions	React (Basic) It covers topics like Basic Routing, Rendering Elements,State Management (Internal Component State), Handling Events, ES6 and JavaScript and Form Validation.
14	SQL (Basic)	30 mins assessment	Solve 2 questions	It includes simple queries, relationships, and aggregators.
15	SQL (Intermediate)	35 mins assessment	Solve 2 questions	It includes complex joins, unions, and sub-queries.
16	SQL (Advanced)	1 hr assessment	Solve 2 questions	It covers topics like query optimization, data modeling, Indexing, window functions, and pivots in SQL.
17	Angular (Basic) Certification	1 hr assessment	Solve 2 questions	It covers topics like MVC Frameworks, Components (Angular, Dynamic, Styling), TypeScript, Two Way Binding and Form Validation
18	Node.js (Intermediate)	1 hr assessment	Solve 2 questions	Understand the concepts behind an Event-Driven Architecture and Concurrency
19	R (Basic)	1 hr 30 mins assessment	Solve 2 questions	This competency area includes fundamentals of the R programming language, understanding Data Frames, Packages, and Data Reshaping, using Data interfaces, among others.
20	Rest API (Intermediate)	1 hr 30 mins assessment	Solve 2 questions	It covers topics like getting data from an API and process using parameters or paging.

IBM (List of Certification Courses) by Class Central /Coursera

https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747

SNo	Topic	Duration	Module	Free/paid
1	IBM: Introduction to Web Development with HTML, CSS, JavaScript	13 hours 3 minutes	Introduction to Application Development HTML Overview CSS Overview & HTML5 Elements JavaScript Programming for Web Applications Career Opportunities and Final Project	Free
2	IBM- Machine Learning with Python	13 hours 24 minutes	**Introduction to Machine Learning** In this module, you will learn about applications of Machine Learning in different fields such as health care, banking, telecommunication, and so on. You'll get a general overview of Machine Learning topics such as supervised vs unsupervised learning, and the usage of each algorithm. Also, you understand the advantage of using Python libraries for implementing Machine Learning models. **Regression** In this module, you will get a brief intro to regression. You learn about Linear, Non-linear, Simple and Multiple regression, and their applications. You apply all these methods on two different datasets, in the lab part. Also, you learn how to evaluate your regression model, and calculate its accuracy. **Classification** In this module, you will learn about classification technique. You practice with different classification algorithms, such as KNN, Decision Trees, Logistic Regression and SVM. Also, you learn about pros and cons of each method, and different classification accuracy metrics. **Linear Classification** **Clustering** In this module, you will learn about clustering specifically k-means clustering. You learn how the k-means clustering algorithm works and how to use k-means clustering for customer segmentation. **Final Exam and Project** In this module, you will do a project based of what you have learned so far. You will submit a report of your project for peer evaluation.	Free
3	IBM- Python for Data Science, AI & Development	1 day 1 hour 3 minutes	**Python Basics** This module teaches the basics of Python and begins by exploring some of the different data types such as integers, real numbers, and strings. Continue with the module and learn how to use expressions in mathematical operations, store values in variables, and the many different ways to manipulate strings. **Python Data Structures** This module begins a journey into Python data structures by explaining the use of lists and tuples and how they are able to store collections of data in a single variable. Next learn about dictionaries and how they function by storing data in pairs of keys and values, and end with Python sets to learn how this type of collection can appear in any order and will only contain unique elements. **Python Programming Fundamentals** This module discusses Python fundamentals and begins with the concepts of conditions and branching. Continue through the module and learn how to implement loops to iterate over sequences, create functions to perform a specific task, perform exception handling to catch errors, and how classes are needed to create objects. **Working with Data in Python** This module explains the basics of working with data in Python and begins the path with learning how to read and write files. Continue the module and uncover the best Python libraries that will aid in data manipulation and mathematical operations. **APIs and Data Collection** This module delves into the unique ways to collect data by the use of APIs and web scraping. It further explores data collection by explaining how to read and collect data when dealing with different file formats.	Free

IBM (List of Certification Courses) by Class Central /Coursera

https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747

SNo	Topic	Duration	Module	Free/paid
4	Machine Learning with Python	3 hours	**Module 1 - Supervised vs Unsupervised Learning :** Machine Learning vs Statistical Modelling, Supervised vs Unsupervised Learning, Supervised Learning Classification Unsupervised Learning **Module 2 - Supervised Learning I :** K-Nearest Neighbors, Decision Trees, Random Forests, Reliability of Random Forests, Advantages & Disadvantages of Decision Trees **Module 3 - Supervised Learning II :** Regression Algorithms, Model Evaluation, Model Evaluation: Overfitting & Underfitting, Understanding Different Evaluation Models **Module 4 - Unsupervised Learning :** K-Means Clustering plus Advantages & Disadvantages, Hierarchical Clustering plus Advantages & Disadvantages, Measuring the Distances Between, Clusters - Single Linkage Clustering, Measuring the Distances Between Clusters - Algorithms for Hierarchy Clustering, Density-Based Clustering **Module 5 - Dimensionality Reduction & Collaborative Filtering :** Dimensionality Reduction: Feature Extraction & Selection, Collaborative Filtering & Its Challenges	Free
5	Introduction to Artificial Intelligence (AI)	8 hours 8 minutes	**What is AI? Applications and Examples of AI** This week, you will learn what AI is. You will understand its applications and use cases and how it is transforming our lives. **AI Concepts, Terminology, and Application Areas** This week, you will learn about basic AI concepts. You will understand how AI learns, and what some of its applications are. **AI: Issues, Concerns and Ethical Considerations** AI is everywhere, transforming the way we live, work, and interact — that's why it's so important to build and use it in line with ethical expectations. This week, you will learn about issues and concerns surrounding AI, as well as how AI ethics helps practitioners build and use AI responsibly. This information will help you understand AI's potential impacts on society so that you can have an informed discussion about its risks and benefits. **The Future with AI, and AI in Action** This week, you will learn about the current thinking on the future with AI, as well as hear from experts about their advice to learn and start a career in AI. You will also demonstrate AI in action by utilizing Computer Vision to classify images.	Free
6	Python Basics for Data Science	3 weeks, 4-10 hours a week	**Module 1 - Python Basics :** Your first program, Types Expressions and Variables String Operations **Module 2 - Python Data Structures :** Lists and Tuples, Sets Dictionaries **Module 3 - Python Programming Fundamentals :** Conditions and Branching, Loops, Functions Objects and Classes **Module 4 - Working with Data in Python :** Reading files with open, Writing files with open, Loading data with Pandas Working with and Saving data with Pandas **Module 5 - Working with Numpy Arrays :** Numpy 1d Arrays Numpy 2d Arrays	Paid

IBM (List of Certification Courses) by Class Central /Coursera

https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747

SNo	Topic	Duration	Module	Free/paid
7	Developing Back-End Apps with Node.js and Express	13 hours 9 minutes	**Introduction to Server-Side JavaScript** You may already be familiar with JavaScript, but did you know there is a server-side JavaScript framework? Discover how Node.js is used on the back end to develop rich, dynamic web applications by importing external modules and using node package manager (NPM). Get hands-on experience using Node.js to build your own simple web server. **Asynchronous I/O with callback programming** How can you optimize processor time with Node.js? Discover how Node.js operates in a non-blocking manner and how to handle the result from a network call. Explore the advantages and challenges associated with using anonymous callback functions. Use Node.js to parse JavaScript Object Notation (JSON) data from a Hypertext Transfer Protocol (HTTP) message. **Express Web Application Framework** When you use Node.js to build a hypertext transfer protocol (HTTP) server, you will discover that the default Node.js framework is low-level. You can extend its features with third-party packages such as Express. Get hands-on experience with the Express web application framework by creating a web server that listens to incoming requests. You also learn how to implement authorization with Express. Use routing, middleware, and template rendering to extend the Node.js framework and handle dynamic content. **Final Project** In this module, you will complete two related projects: a practice project and a final project. The practice project walks you through code that performs CRUD operations for given user details on an Express server using session and JWT authentication and then tests the code using Postman. For the final project, you assume the role of a back-end developer working on a server-side application that stores, retrieves, and manages book ratings and reviews. You will submit screenshots of your application, and then your peers will review the project based on a rubric provided. You will also be expected to review two of your peer's projects.	Paid
8	IBM Front-End Developer		Course 1: Getting Started with Front-End and Web Development Course 2: Introduction to Software Engineering Course 3: Designing User Interfaces and Experiences (UI/UX) Course 4: Introduction to HTML, CSS, & JavaScript Course 5: Developing Websites and Front-Ends with Bootstrap Course 6: Getting Started with Git and GitHub Course 7: Developing Front-End Apps with React Course 8: Intermediate Web and Front-End Development Course 9: Get Started with Cloud Native, DevOps, Agile, and NoSQL Course 10: Front-End Development Capstone Project Course 11: Software Developer Career Guide and Interview Preparation	Paid
9	IBM Full-Stack JavaScript Developer		Course 1: Introduction to Software Engineering Course 2: Introduction to HTML, CSS, & JavaScript Course 3: Getting Started with Git and GitHub Course 4: JavaScript Programming Essentials Course 5: Developing Front-End Apps with React Course 6: Developing Back-End Apps with Node.js and Express Course 7: Get Started with Cloud Native, DevOps, Agile, and NoSQL Course 8: Introduction to Containers w/ Docker, Kubernetes & OpenShift Course 9: Application Development using Microservices and Serverless Course 10: Node.js & MongoDB: Developing Back-end Database Applications Course 11: JavaScript Full Stack Capstone Project Course 12: Software Developer Career Guide and Interview Preparation	Paid

IBM (List of Certification Courses) by Class Central /Coursera

https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747

SNo	Topic	Duration	Module	Free/paid
10	IBM Full Stack Software Developer	17 weeks, 10 hours a week	Course 1: Introduction to Cloud Computing Course 2: Introduction to Web Development with HTML, CSS, JavaScript Course 3: Getting Started with Git and GitHub Course 4: Developing Front-End Apps with React Course 5: Developing Back-End Apps with Node.js and Express Course 6: Python for Data Science, AI & Development Course 7: Developing AI Applications with Python and Flask Course 8: Django Application Development with SQL and Databases Course 9: Introduction to Containers w/ Docker, Kubernetes & OpenShift Course 10: Application Development using Microservices and Serverless Course 11: Full Stack Application Development Capstone Project Course 12: Full Stack Software Developer Assessment	Paid

Oracle -My Learn Courses

https://mylearn.oracle.com/ou/course/mysql-80-for-developers/81909

SNo	Topic	Duration	Module	Free/paid
1	MySQL 8.0 for Developers	14h 20m	The MySQL for Developers training course teaches developers you how to develop console and web applications using MySQL with your choice of the PHP, Java, or Python programming languages. Expert Oracle University instructors will guide you through realistic hands-on activities in which you will learn how to use Connectors to access MySQL databases, query the database effectively, present data in different formats, and how to use MySQL's support for JSON and spatial data. Learn to: Program with Connectors Write console and web applications Optimize query performance Write stored routines and triggers Store, access, and analyze geospatial data Use X DevAPI to access JSON data in document store Benefits to you This course will teach you how to write applications that maximize your investment in MySQL. You will learn best practice techniques for writing programs that store and retrieve MySQL data and present it to users in a way that allows them to gain insight into that data. Additional Information Version : 8.0 Role : Database Administrator Language : English	Free
2	Oracle Database 19c: SQL Workshop	22 Modules 14h 41m	Benefits to you Learn how to: Retrieve Data using the SQL SELECT Statement Restrict and Sort Data Customize Output Using Single-Row Functions Use Conversion Functions and Conditional Expressions Report Aggregated Data Using the Group Functions Display Data from Multiple Tables Using Joins Manage Tables by Using DML Statements Create Sequences, Synonyms, Indexes, Views, and Schema Objects Manage Data by Using Subqueries and Advanced Queries	Free
3	Prepare for Oracle Database SQL Certification	4 Modules 1h 27m	Benefits to you Prepare for Oracle Database SQL Certification Additional Information Getting Started and New Releases : Exam Preparation Role : Database Developer Learn about Oracle's Products : SQL Version : 19c Language : English	Free

Oracle -My Learn Courses

https://mylearn.oracle.com/ou/course/mysql-80-for-developers/81909

SNo	Topic	Duration	Module	Free/paid
4	Java		Program in Java using Java SE 8 and Java SE 11, the long-term support releases, and most widely used versions of Java SE. Program using Java EE 7, which is the platform for enterprise Java applications. Use the module system to design applications with explicit dependencies and encapsulation at the JAR level, ensuring more reliable configuration, improved security, and enhanced performance. Make your code more readable and succinct using convenience methods. Test code snippets and APIs using JShell. Use Java Persistence and Java Transaction APIs, create a flexible component model using EJB and CDI technology, provide Timer, Concurrency and Batch services, and create SOAP and REST web services. Develop web-based interfaces using Servlets, Java Server Pages, and JavaServer Faces. Implement best practices for writing Java code. Leverage product enhancements soon after each product release. Take advantage of all training in the learning subscription in order to learn and get certified on Java SE 8, Java SE 11, and Java EE 7. Learn to implement WebLogic Server, SOA Cloud, and WebCenter Portal cloud services in creating applications on Oracle Cloud.	Free

Google (List of Certification Courses) by Coursera

https://grow.google/intl/en_in/certificates/

SNo	Topic	Lavel	Duration	Timing	Outcomes	Module	Recommended for	Fee
1	Python for Data Science, AI & Development	Beginner level	25 hours (approximately)	Flexible schedule	Learn new concepts from industry experts Gain a foundational understanding of a subject or tool Develop job-relevant skills with hands-on projects Earn a shareable career certificate	Python Basic Python Data Structure Python Programming Fundamental Working with Data in Python API and Data Collection	Recommended if you're interested in Data Analysis	Free
2	Introduction to Microsoft Excel	Intermediate level	1.5 hours		Learn step-by-step In a video that plays in a split-screen with your work area, your instructor will walk you through these steps: • The Microsoft Office365 Excel Program and view the basics of Excel (6 min) • Learn the layout, navigation, and ribbon (7 min) • Create a spreadsheet of information you are tracking (5 min) • Create basic formulas to analyze your data (6 min) • Format your data (6 min)			Free
3	Python for Everybody Specialization	Beginner level	2 months at 10 hours a week	Flexible schedule	Learn in-demand skills from university and industry experts Master a subject or tool with hands-on projects Develop a deep understanding of key concepts Earn a career certificate from University of Michigan	Programming for Everybody (Getting Started with Python) Python Data Structures Using Python to Access Web Data Using Databases with Python Capstone: Retrieving, Processing, and Visualizing Data with Python		Free

Google (List of Certification Courses) by Coursera

https://grow.google/intl/en_in/certificates/

SNo	Topic	Lavel	Duration	Timing	Outcomes	Module	Recommended for	Fee
4	Programming in Python	Beginner level	44 hours (approximately)	Flexible schedule	This course is available as part of When you enroll in this course, you'll also be asked to select a specific program. Learn new concepts from industry experts Gain a foundational understanding of a subject or tool Develop job-relevant skills with hands-on projects Earn a shareable career certificate from Meta	Get started with the Python programming language and associated foundational concepts. Learn basic Python syntax, to use control flow and loops and work with functions and data structures. You will also learn how to recognise possible errors, their causes and how to handle them. Learn about the paradigms of procedural programming and the associated logical concepts. You'll explore functional and object-oriented programming, and get an introduction to algorithms. Supercharge your coding environment with popular modules libraries and tools for Python. You'll also learn about the different types of testing and how to write a test. Practice and reflect on the skills you learned in this course.		
5	HTML, CSS, and Javascript for Web Developers		8 hours (approximately)	Flexible schedule	This course is part of the HTML, CSS, and Javascript for Web Developers Specialization When you enroll in this course, you'll also be enrolled in this Specialization. Learn new concepts from industry experts Gain a foundational understanding of a subject or tool Develop job-relevant skills with hands-on projects Earn a shareable career certificate	In this course we will learn the basics of HTML5. We'll start with instructional videos on how to set up your development environment, go over HTML5 basics like valid document structure, which elements can be included inside other elements and which can not, discuss the meaning and usefulness of HTML5 semantic tags, and go over essential HTML5 tags.	Recommended if you're interested in Mobile and Web Development	free

Oracle Certifications

https://mylearn.oracle.com/ou/home

Oracle Certification levels	Foundations	**Foundations** An Oracle Certified Foundations associate has the foundational knowledge of core concepts and services. These are intended for candidates with technical and non-technical backgrounds who want to validate their understanding of Oracle technologies. **Prepares you for these roles** Administrator Implementer Developer Recognizes these skills Focused on Foundational level knowledge and skills Foundations level exams require an understanding of the concepts and fundamentals
	Associate	**Associate** An Oracle Certified Associate has a technical background and a strong understanding of Oracle technologies. An Associate certification is intended for candidates who can use their knowledge and experience to apply Oracle's recommended best practices in the respective domain. **Prepares you for these roles** Administrator Implementer Developer Architect Data scientist/business analyst Recognizes these skills Foundational and intermediate knowledge and skills
	Professional	**Professional** An Oracle Certified Professional is skilled in Oracle technologies. A Professional certification validates advanced understanding of Oracle concepts and skills as well as experience applying these in real-world scenarios. **Modules** Administrator Implementer Developer Consultant Architect Recognizes these skills Advanced skills and strong foundational knowledge of the technology

SNo	Topic	Test Duration	Exam Pattern	Exam Topics
1	MySQL 8.0 Database Developer Exam Number: 1Z0-909	90 Minutes	Number of Questions: 65 Passing Score: 62% Validation: Exam has been validated for product version MySQL 8.0.18	Review exam topics Connectors and APIs Choose between connectors for a given application, Demonstrate connector use, management and configuration, Retrieve data from the database by using a connector, Handle special values, Secure credentials and connections, Data-driven Applications Use prepared statements, Set SQL Modes to change MySQL behavior, Handle and interpret errors and warnings, Display query results, Aggregate and summarize data, Advanced Report Generation, MySQL Schema Objects and Data Design, create, and alter views, Store and process string data, Store and process numeric data, Store and process temporal data, Store and process spatial data, Transactions Control transactions in SQL, Control transactions in applications, Resolve consistency problems with isolation levels, Understand locking mechanisms within MySQL, Query Optimization Use indexes for optimization, Analyze queries for optimization, Rewrite queries for optimization, MySQL Stored Programs, Create and execute stored routines, Explain the programming constructs used in stored routines, Create and execute triggers, Schedule database operations, JSON and Document Store Create and store JSON documents, Process data in JSON documents, Explain application development with NoSQL and XDevAPI, Create and access a document store, Use MySQL Shell to access document stores
2	Oracle Database SQL 1Z0-071	2Hr.	---	-----
3	Exam: Oracle Java Foundations (1Z0-811-A)	180 Minutes	---	-----

TCS iON - PAID CERTIFICATE COURSES (Technical)

https://www.tcsion.com/hub/career-creator/free-learning-courses/

SNo	Course	Session	Course Format	Who would benefit?	Pre-requisite	Syllabus	Fee Amount
1	Learn PHP and MySQL from Scratch	Duration: 5 Weeks Commitment: 10 Hours/Week	Online Self-paced	A graduate or postgraduate student, wanting to learn this skill		Module 1 Introduction to PHP and Source Code, Module 2 PHP Variables and Data Types, Module 3 String Functions, Module 4 Conditional and Looping, Module 5 Functions, Module 6 GET and POST, Module 7 Array Functions, Module 8 PHP Emails and Filtering Inputs, Module 9 Activity: Create a Responsive Contact Form Using PHP and Bootstrap, Module 10 Date and Time, Module 11 Files and File Handling, Module 12 Uploading Files, Module 13 Cookies, Module 14 Error Handling, Module 15 Sessions, Module 16 Test Your Knowledge, Module 17 Introduction to MySQL and Source Code, Module 18 PHP My Admin and Remote MySQL, Module 19 Use PHP and MySQL to Create/Populate Databases and Tables, Module 20 Activity: Populate Database Once the User Submits a Form, Module 21 Activity: Populate an HTML Table Using Database Data, Module 22 Update/Delete Data, Module 23 Test Your Knowledge	Rs. 977/-
2	Introduction to Artificial Intelligence with Java	"Duration: 1 Weeks Commitment: 2-3 Hours"	Online Self-paced	A graduate or postgraduate student, wanting to learn this skill		Module 1 Introduction to Artificial Intelligence and Java Module 2 Exploring Search Module 3 AI Games and Rule-Based System Module 4 Interfacing with Weka Module 5 Handling Attributes Module 6 Supervised Learning Module 7 Semi-Supervised and Unsupervised Learning	Rs. 1115/-
3	Learn HTML from Scratch	Duration: 1 Weeks Commitment: 6 Hours	Online Self-paced	A graduate or postgraduate student, wanting to learn this skill	No. There are no prerequisites to complete this course.	Module 1 Introduction and Source Code Module 2 Your First Website Module 3 Set Up Your Free Hosting with FireFTP Module 4 Headings, Paragraphs, Links and Images Module 5 Inline vs Block Elements Module 6 Iframes - Activity: Embed a Nice Relaxing YouTube Video to Your Website Module 7 Lists Module 8 Tables, Entities and Forms Module 9 Text Decoration and Comments Module 10 Test Your Knowledge	Rs. 977/-
4	Build an Online Notes App (HTML, CSS, Bootstrap, PHP, MySQL and AJAX)	Duration: 3 Weeks Commitment: 10 Hours/ Week	Online Self-paced	A graduate or postgraduate student, wanting to learn this skill	Basic knowledge of HTML, CSS, JavaScript, PHP and MySQL is required for taking up this course.	Module 1 Project Introduction and Source Code Module 2 App Skeleton and Decoration Module 3 App Logic - Signup, Login, Remember Me, Forgot Password Module 4 Signup/Login Code Structure Module 5 Signup/Login Code Module 6 Remember Me Module 7 Forgot Password Code Module 8 Notes Management: Logic and Code Structure Module 9 Load, Create, Edit and Delete Notes Module 10 Update Username, Password and Email	Rs. 977/-

TCS iON - PAID CERTIFICATE COURSES (Technical)

https://www.tcsion.com/hub/career-creator/free-learning-courses/

SNo	Course	Session	Course Format	Who would benefit?	Pre-requisite	Syllabus	Fee Amount
5	Build a Fruits Slice Game (HTML, CSS and jQuery)	Duration: 1 Weeks Commitment: 4 Hours	Online Self-paced	A graduate or postgraduate student, wanting to learn this skill	Basic knowledge of HTML, CSS, JavaScript and jQuery is required for taking up this course.	Module 1 Project Introduction Module 2 Page Structure and Styling Module 3 Game Logic Module 4 jQuery Code Module 5 Source Code	Rs. 977/-

TCS iON - PAID ASSESSEMENT

https://www.tcsion.com

SNo	Test Name	Next Test Date	Upcoming Test Date	About the Test	How Can Apply	Part A - Test of knowledge	Syllabus	Amount/ Fee
1	**TCS iON NPT - JavaScript Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to certify their knowledge and skills in JavaScript Development. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - JavaScript Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - JavaScript Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles related to JavaScript Development in top corporates	Variables, Types, and Collections: JavaScript, knowledge of JS variables, all new features in ES6 Objects, Functions, and Classes: Object-Oriented Programming and its principles, Functions and their types and more Browser and Events: Working of Web Browser, Integrating JavaScript to web browser, DOM Parsing and Event Handling and more Debugging and Error Handling: Errors, and Exception Handling Asynchronous Programming and Server-side JavaScript: JavaScript, AJAX, knowledge of Server-side JavaScript Testing: Testing techniques, their tools and test scenarios Data Storage and Security: Client-side data storage	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTJSD2232023/microsite_1715834844520/pdf/View%20Syllabus.pdf?version=1854575300 Questions - 50 Marks - 50 Time - 60 Mins	**Rs. 799/-**
2	**TCS iON NPT - Android Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Android Development. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Android Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Android Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles related to Android Development in top corporates	Fundamentals of Kotlin: Knowledge of Kotlin Fundamentals Working with Android Framework and Android Studio: Android Framework, Android Studio and Android SDK Creating User Interface: User Interface design in Android, Various Layouts in Android, Styling and Menu creation in Android Firebase Authentication and Database: Server-side operations, APIs, Database Operations, Firebase Location-Aware Apps: Location-Aware Apps, GPS and Google Maps App Testing and Publishing: Testing and publishing the Android Application **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTANDT22032023/microsite_1715751103951/pdf/View%20Syllabus.pdf?version=1854574523	**Rs. 799/-**
3	**TCS iON NPT - MEAN Stack Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in MEAN Stack Development. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - MEAN Stack Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - MEAN Stack Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles related to MEAN Stack Development in top corporates	Basics, Installation and Setup: MEAN Stack Development Basics, Installation of various tools and configuring them, setting up the development environment Building User Interfaces with Angular: Front-end operations, AngularJS basics and Designing the UI with Angular Building Server-side with Node.js: Server-side Scripting, basics of Node.js, Required Modules and Components, NPM, HTTP Servers and more Building APIs with Express: Server-side operations, APIs, Express JS Managing Date with MongoDB: Database, Database Servers, NOSQL Database, MongoDB Database Putting it all together: Integration of front-end, back-end and APIs and building a high-end web application **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTMSD2332023/microsite_1715754888216/pdf/View%20Syllabus.pdf?version=1854575375	**Rs. 799/-**

TCS iON - PAID ASSESSEMENT

https://www.tcsion.com

SNo	Test Name	Next Test Date	Upcoming Test Date	About the Test	How Can Apply	Part A - Test of knowledge	Syllabus	Amount/ Fee
4	**TCS iON NPT - Python Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Python Programming. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Python Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Python Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles in Python Programming in top corporates	Statistics: Basics of Statistics Core Python: Core Python with Object-oriented programming NumPy: NumPy for N-dimensional arrays Data Munging: Data munging in Python using Pandas EDA: Exploratory Data Analysis in Python using Pandas Matplotlib: Visualisation using Matplotlib	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTPDPT232022/microsite_1715832966750/pdf/View%20Syllabus.pdf?version=1854574175	**Rs. 799/-**
5	**TCS iON NPT - Java Full Stack Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Java Full Stack Development. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Java Full Stack Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Java Full Stack Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and working professionals who would like to prove their skills in Java Full Stack Development Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles in Java Full Stack Development	Front-end Technology: To test the knowledge of HTML5 and CSS3, all new features in web technology, basic understanding of Javascript, jQuery Library, Javascript Frameworks like AngularJS, ReactJS, and more Design Ability: Basic prototype design and UI/UX design Version Control System(VCS): To check the knowledge of Version Control Systems like GIT, their commands and operations Server: To verify the basic knowledge of networking fundamentals, servers and server configurations Testing: To check the knowledge of Unit testing and DSA Development Language: To check the knowledge of Programming Constructs, OOP, JSP, Spring MVC Implementation, JDBC, and more **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTJAVAFS09032022/microsite_1715584309852/pdf/View%20Syllabus.pdf?version=1854575437	**Rs. 799/-**
6	**TCS iON NPT - IT Web Designer**	**6th November, 2024**	To be Announced	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in IT Web Designing. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - IT Web Designer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - IT Web Designer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles related to IT Web Designing in top corporates	HTML Building Components - Basic HTML controls and structure New Features of HTML5 - Handling HTML5 complex components CSS Properties - Web application look and feel will be handled by CSS properties and Knowledge of CSS Properties CSS Component - CSS Properties can be applied to the HTML elements. CSS component helps programmer to perform actions on HTML components. Helps in enhancing application performance Java - Focussed on back end service implementations. Core Java, Advanced Java and Framework knowledge introduced in this section DevOps - Application Deployment Process, Configuration, Testing and Performance Management incorporated **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNQTITWEBD01112021/microsite_1715752273128/pdf/View%20Syllabus.pdf?version=1854573666	**Rs. 799/-**

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SNo	Test Name	Next Test Date	Upcoming Test Date	About the Test	How Can Apply	Part A - Test of knowledge	Syllabus	Amount/ Fee
7	**TCS iON NPT - Artificial Intelligence**	**6th November, 2024**	To be Announced	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Artificial Intelligence (AI). All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Artificial Intelligence Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Artificial Intelligence Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for Artificial Intelligence (AI) related job roles in top corporates Get Started	Introduction to Artificial Intelligence - Basic concepts of Artificial Intelligence (AI) Search - Focusses on search algorithms which forms the base of Artificial Intelligence (AI) Reasoning - Focusses on reasoning issues like logical reasoning which is beneficial from Artificial Intelligence point of view Machine Learning - Focusses on Machine Learning concepts and types of Machine Learning Deep Learning - Deep Learning is a section of Machine Learning. This module focusses on the concepts of Deep Learning Time Series - Time series is a pattern which is commonly followed in Artificial Intelligence (AI). This section focusses on implementation of Time Series TensorFlow - TensorFlow is a Deep Learning framework created by Google and this module focusses on the basics and terminologies of Tensorflow **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNQTARTINT240621/microsite_1715751826256/pdf/View%20Syllabus.pdf?version=1854577370	**Rs. 799/-**
8	**TCS iON NPT - Machine Learning**	**6th November, 2024**	To be Announced	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Machine Learning. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Machine Learning Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Machine Learning Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for Machine Learning related job roles in top corporates	Basics of Statistics - Basic concepts of Statistics Basics of Machine Learning - Focusses on the terminologies of Machine Learning and Algorithms Machine Learning Methodology (CRISP DM) - Covers all methodologies of Machine Learning Key concepts of Machine Learning - Covers all the concepts of Machine Learning Machine Learning Algorithms with real-life use cases - Covers all the real-life use cases Exploratory Data Analysis - Exploratory data analysis (EDA) is an approach to analysing data sets to summarise their main characteristics, often with visual methods **Questions - 50 Marks - 50 Time - 60 Mins** Part B - Test of application, Data Wrangling and EDA, Python for Machine Learning, Supervised Machine Learning Algorithms with real life use cases	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNQTMLEARN230621/microsite_1715772429944/pdf/View%20Syllabus.pdf?version=1854577445	**Rs. 799/-**
9	**TCS iON NPT - Basic Computer Skills**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge of Basic Computer Skills. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Those candidates who secure a score of 60% and above, will be provided a certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Anyone aspiring for job roles that require Basic Computer Skills, such as sending emails, documentation, research on the internet using browsers and other related tasks	Basic Hardware/Software Components - Skills to use computer systems properly for day-to-day activities/tasks, Microsoft Word - Skills to create formatted/creative documents/reports using word processing software, Microsoft Excel - Skills to create worksheets, analyse data and visualise results using spreadsheet software, Microsoft PowerPoint-MS-PowerPoint Basics, Slide Formating, Making Presentations, Computer Network & Internet, Web Browsers & Emailing **Questions - 40, Marks - 40, Time - 40 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNQTBASICCOMSKILL01022021/microsite_1715832183937/pdf/View%20Syllabus.pdf?version=1854575902	**Rs. 399/-**

CISCO (List of Certification Courses)

https://www.cisco.com/site/us/en/learn/training-certifications/index.html

S.No.	NAME OF COURSES	CERTIFICATION/ DIGITAL BADGES	LENGTH OF COURSE	COST	DESCRIPTION (SKILLS)	LEVEL	LEARNING TYPE
1	Networking Essentials	DIGITAL BADGE	70 hours	Free*	Core Skills: Plan and install a home or small business network using wireless technology, then connect it to the Internet. Develop critical thinking and problem-solving skills using Cisco Packet Tracer. Practice verifying and troubleshooting network and Internet connectivity. Recognize and mitigate security threats to a home network.	Intermediate	Instructor-led, Online self-paced
2	Introduction to Cybersecurity	DIGITAL BADGE	15 hours	Free*	Core Skills: Learn what cybersecurity is and its potential impact to you. Understand the most common threats, attacks and vulnerabilities. Gain insights for how businesses protect their operations from attacks. Find the latest job trends and why cybersecurity continues to grow.	Beginning	Instructor-led, Online self-paced
3	Cybersecurity Essentials	DIGITAL BADGE	30 hours	Free*	Core Skills: Understand security controls for networks, servers and applications. Learn valuable security principals and how to develop compliant policies. Implement proper procedures for data confidentiality and availability. Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer.	Intermediate	Instructor-led, Online self-paced
4	CyberOps Associate	Badge, Certification Aligned	70 hours	Determined by Institution	Core Skills: Deepen knowledge in how best to detect and respond to security incidents. Gain job-ready, practical skills in cybersecurity operations. Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Prepare for Cisco Certified CyberOps Associate Certification.	Intermediate	Instructor-led,
5	Introduction to IoT	DIGITAL BADGE	20 hours	Free*	Core Skills: Learn how the current digital transformation is creating unprecedented economic opportunity. Understand how the IoT is bridging the gap between operational and information technology systems. Discover how standard business processes are being transformed. The security concerns that must be considered when implementing IoT solutions.	Beginning	Instructor-led, Online self-paced
6	IoT Fundamentals: Connecting Things	--------	70 hours	Determined by Institution	Core Skills: Learn how the current digital transformation is creating unprecedented economic opportunity. Understand how the IoT is bridging the gap between operational and information technology systems. Develop critical thinking and problem-solving skills using both real equipment and Cisco Packet Tracer. Soft skills such as teamwork and articulating problems and solutions in a business context.	Intermediate	Instructor-led,

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S.No.	NAME OF COURSES	CERTIFICATION/ DIGITAL BADGES	LENGTH OF COURSE	COST	DESCRIPTION (SKILLS)	LEVEL	LEARNING TYPE
7	IoT Fundamentals: Big Data & Analytics	--------	50 hours	Determined by Institution	Core Skills: Use Python to create code that reads data from sensors and stores it in a SQL database. Visualize, clean, manipulate and integrate data sets. Learn fundamental principles of Big Data platforms like Hadoop. Use storytelling to present insights gained from extracted data.	Intermediate	Instructor-led,
8	IoT Fundamentals: Hackathon Playbook	--------	30 hours	Determined by Institution	Core Skills: Identify a problem, create a design, then plan for an IoT solution. Develop a prototype of an IoT solution. Test and validate your prototype. Present IoT concept and solution to experts.	Intermediate	Instructor-led,
9	Get Connected	--------	30 hours	Free*	Core Skills: Create accounts for Facebook, LinkedIn and YouTube. Understand computer networking and how to browse and search the Internet. Become familiar with Microsoft Windows and how to work with files and folders. Identify common problems and implement solutions.	Beginning	Instructor-led, Online self-paced
10	IT Essentials	Badge, Certification Aligned	70 hours	Determined by Institution	Core Skills: Install, configure, and troubleshoot computers and mobile devices. Identify common security threats like phishing and spoofing. Develop critical thinking and problem-solving skills using both real equipment and Cisco Packet Tracer. Prepare for CompTIA A+ Certification.	Intermediate	Instructor-led,
11	NDG Linux Unhatched	--------	08 hours	Free*	Core Skills: Basic installation and configuration of Linux software. Understand the basics of the Linux Command Line Interface (CLI). Interact with the Linux virtual machine. Determine if Linux is for you or not.	Beginning	Online self-paced
12	NDG Linux Essentials	--------	70 hours	Free*	Core Skills: Acquire open source concepts and progressively master Linux commands. Understand how Linux is used and the basics of its command line. Become skilled at using the Linux virtual machine for experiments. Prepare for LPI Linux Essentials Professional Development Certificate (PDC).	Intermediate	Instructor-led, Online self-paced

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S.No.	NAME OF COURSES	CERTIFICATION/ DIGITAL BADGES	LENGTH OF COURSE	COST	DESCRIPTION (SKILLS)	LEVEL	LEARNING TYPE
13	NDG Linux I	--------	70 hours	$29.95 for self-paced classes; cost for instructor-led classes is determined by the institution	You'll Learn These Core Skills: Gain a working knowledge of the Linux command line. Install and configure a computer running Linux. Configure basic networking using virtual machines. First course in 2-course series needed for preparing for LPIC-1 (Exam 101) Certification.	Advanced	Instructor-led, Online self-paced
14	NDG Linux II	Certification Aligned	70 hours	$29.95 for self-paced classes; cost for instructor-led classes is determined by the institution	Core Skills: Augment your basic knowledge to become skilled in the Linux command line. Configure basic networking, using virtual machines running Linux. Perform scripting and data management; learn about interfaces, networking fundamentals and security. Prepare for certification LPIC-1 (Exam 102).	Advanced	Instructor-led, Online self-paced
15	PCAP: Programming Essentials in Python	Certification Aligned	75 hours	Free*	core skills: Think algorithmically – how to analyze a problem and translate it for a computer to process. Design, develop, and improve multi-module computer programs. Analyze and model real-life problems in Object-Oriented Programming categories. Understand a programmer's work in the software development process. Learn how a program is executed in a computer environment. Gain skills to create and develop your own programming portfolio.	Intermediate	Instructor-led, Online self-paced
16	CLA: Programming Essentials in C	Certification Aligned	70 hours	Determined by Institution	Core Skills: Understand common computer programming concepts and write your own programs. Learn the syntax, semantics and basic data types of C. Apply programming skills using hands-on lab activities. Prepare for the C Programming Language Certified Associate Certification (CLA).	Intermediate	Instructor-led,
17	CPA: Programming Essentials in C++	Certification Aligned	70 hours	Free*	Core Skills: Learn the syntax, semantics, and basic data type of C++. Understand principles of the object-oriented model. Implement and write in C++ and resolve typical implementation challenges via language libraries. Prepare for the C++ Certified Associate Programmer Certification (CPA).	Intermediate	Instructor-led, Online self-paced
18	CPP: Advanced Programming in C++	Certification Aligned	70 hours	Determined by Institution	Core Skills: Learn the C++ template mechanism and write your own C++ programs. Understand and use elements of the C++ STL library. Solve programming problems with STL-predefined classes and methods. Prepare for the C++ Certified Professional Programmer Certification (CPP).	Intermediate	Instructor-led,

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S.No.	NAME OF COURSES	CERTIFICATION/ DIGITAL BADGES	LENGTH OF COURSE	COST	DESCRIPTION (SKILLS)	LEVEL	LEARNING TYPE
19	JavaScript Essentials 1 (JSE)	Certification Aligned	40 hours	Free*	core skills: Think algorithmically and analyze problems. Design, develop, and improve JavaScript programs. Understand a programmer's work in the software development process and the role of fundamental development tools. Know how a program is interpreted and executed in an actual local and remote computer environment. Create and develop your own programming portfolio.	Beginning	Instructor-led, Online self-paced
20	CCNA: Introduction to Networks	DIGITAL BADGE	70 hours	Determined by Institution	Core Skills: Build simple LANs, perform basic configurations for routers and switches, and implement IPv4 and IPv6 addressing schemes. Configure routers, switches, and end devices to provide access to local and remote network resources and to enable end-to-end connectivity between remote devices. Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Configure and troubleshoot connectivity a small network using security best	Intermediate	Instructor-led,
21	Switching, Routing, and Wireless Es	--------	70 hours	Determined by Institution	Core Skills: Work with routers, switches and wireless devices to configure and troubleshoot VLANs, Wireless LANs and Inter-VLAN routing. Configure and troubleshoot redundancy on a switched network using STP and EtherChannel. Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Explain how to support available and reliable networks using dynamic addressing and first-hop redundancy protocols.	Intermediate	Instructor-led,
22	DevNet Associate	Badge, Certification Aligned	70 hours	Determined by Institution	Core Skills: Practice software development skills, including Python, GIT, and common data formats (XML, JSON, and YAML). Become familiar with infrastructure automation using code, DevOps methodology, and microservices. Deploy applications as containers and use Continuous Integration/Continuous Deployment (CI/CD) pipelines. Develop critical thinking and problem-solving skills using Cisco Packet Tracer and virtual machines.	Intermediate	Instructor-led,
23	IoT Fundamentals: IoT Security	--------	50 hours	Determined by Institution	Core Skills: Conduct end-to-end security assessments of IoT systems to demonstrate vulnerabilities. Gain hands-on experience with IoT prototypes using a Raspberry Pi. Recommend threat mitigation measures to minimize the risk in IoT solutions and networks. Become proficient using real-world penetration and vulnerability testing tools such as Kali Linux.	Intermediate	Instructor-led,
24	Workshop: Model-Driven Programmability	DIGITAL BADGE	08 hours	Determined by Institution	Core Skills: Use Python with RESTCONF and NETCONF APIs to retrieve and update the device's configuration. Describe a different approach to software-defined networking (SDN), including central application policy control. Use software concepts and tools relevant to network programmability including Python scripting, Git, JSON, Postman, and APIs. Join developer professional communities such as Cisco DevNet, GitHub, and Stack Overflow.	Intermediate	Instructor-led,

CISCO (List of Certification Courses)

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S.No.	NAME OF COURSES	CERTIFICATION/ DIGITAL BADGES	LENGTH OF COURSE	COST	DESCRIPTION (SKILLS)	LEVEL	LEARNING TYPE
25	Workshop: REST APIs using Webex Teams	--------	08 hours	Determined by Institution	Core Skills: Move from hearing buzzwords to having real hands-on experience with how "REST APIs" support the "Programmability" of applications. Learn why Python is the high-demand software language for network programmability, and use it with developer tools like JSON and Postman. Develop first application on a real-world cloud collaboration platform with Webex Teams. Join developer professional communities such as Cisco DevNet, GitHub, and Stack Overflow.	Intermediate	Instructor-led,
26	CCNA: Enterprise Networking, Security, and Automation	Badge, Certification Aligned	70 hours	Determined by Institution	Core Skills: Work with routers and switches using OSPF in point-to-point and multiaccess networks. Mitigate threats and enhance network security using access control lists and security best practices. Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Understand virtualization, SDN, and how APIs and configuration management tools enable network automation.	Advanced	Instructor-led,
27	CCNP Enterprise: Advanced Routing	Badge, Certification Aligned	70 hours	Determined by Institution	Core Skills: Deploy advanced IPv4 and IPv6 technologies to support a scalable enterprise network architecture. Configure and manage secure site-to-site and remote-access networks using tunneling technologies. Optimize network performance using multi-protocol route redistribution and conditional forwarding. Implement secure mechanisms for network infrastructure and troubleshoot security threats.	Advanced	Instructor-led,
28	CCNP Enterprise: Core Networking	--------	70 hours	Determined by Institution	Core Skills: Implement advanced technologies to support a secure and scalable enterprise network architecture. Configure enterprise networks for high availability and optimized performance. Configure and manage secure site-to-site, remote-access, and wireless networks. Deepen your understanding of virtualization and network automation.	Advanced	Instructor-led,
29	CLP: Advanced Programming in C	Certification Aligned	70 hours	Determined by Institution	Core Skills: Grow your knowledge of the syntax and semantics of the C language. Design and write C programs regardless of the hardware of software platform. Identify code bugs and solve complex problems using data structures and algorithms. Prepare for vendor-neutral C Certified Professional Programmer (CLP) Certification.	Advanced	Instructor-led,
30	Network Security	DIGITAL BADGE	70 hours	Determined by Institution	Core Skills: Develop an in-depth understanding of network security. Design, implement, and support security for networked devices and data. Earn critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Gain industry recognized skills aligned with the National Institute for Standards and Technology (NIST) Cybersecurity Framework.	Advanced	Instructor-led,

Saylor Academy, USA (List of Certification Courses)

https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start

S.No.	NAME OF COURSES	certification available?	Duration	Cost	Eligibility for certification	Remarks
	Domain : COMPUTER SCIENCE					
1	CS101: Introduction to Computer Science I	Certificate Course	52 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Explore the basic concepts, nomenclature, and historical perspective of computers and computing, and principles of software development and Object-Oriented Programming.** This course will introduce you to the field of computer science and the fundamentals of computer programming. CS101 is specifically designed for students with no prior programming experience and touches upon a variety of fundamental topics. This course uses Java to demonstrate those topics. Java is a high-level, portable, and well-constructed computer programming language developed by Sun Microsystems (now Oracle).
2	CS102: Introduction to Computer Science II	Certificate Course	51 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Explore this detailed survey of computing and programming, with an emphasis on understanding object-orientation and the Java and C++ computer programming languages. We will use history, theory, and practice to deliver lessons that prepare you for a career in computer science.** We will begin with a comparison between Java, the programming language used in the previous course, and C++, another popular, industry-standard programming language. We will then discuss the fundamental building blocks of Object-Oriented Programming, reviewing what we have already learned while familiarizing ourselves with more advanced programming concepts. The remaining course units will be devoted to various topics, including the Standard Template Library, containers, exceptions, recursion, searching and sorting, and generic programming. By the end of the class, you will have a solid understanding of Java and C++ programming, as well as a familiarity with the major issues that programmers routinely address in a professional setting.
3	CS105 Introduction to Python	Certificate Course	36 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn fundamental programming concepts using the Python 3 programming language, a high-level interpreted language that is easy to read write, with powerful libraries that provide additional functionality.** This course is an introduction to fundamental programming concepts by way of the Python 3 programming language. Python 3 is a high-level interpreted language that has many benefits, including easy-to-read and easy-to-write syntax and powerful libraries that provide additional functionality. Even though Python 3 is a great programming language for beginners, it is also used extensively for practical applications in engineering and data science. This course is intended for people with no or very little prior programming experience. It covers a range of topics, such as data types, control flow, functions, file operations, and object-oriented programming. When you finish this course, you will be able to create Python programs for a variety of applications.
4	CS107 C++ Programming	Certificate Course	40 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn the C++ computer programming language, with a focus on syntax for primitive types, control structures, vectors, strings, structs, classes, functions, file I/O, exceptions, and other programming constructs.** In this course, we will learn the mechanics of editing and compiling programs in C++. We will begin with a discussion of the essential elements of C++ programming: variables, loops, expressions, functions, and the string class. Then, we will cover the basics of object-oriented programming: classes, inheritance, templates, exceptions, and file manipulation. We will then review function and class templates and the classes that perform output and input of characters to/from files. This course will also cover namespaces, exception handling, and preprocessor directives. In the last part of the course, we will learn some slightly more sophisticated programming techniques that deal with data structures such as linked lists and binary trees.

Saylor Academy, USA (List of Certification Courses)

https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start

S.No.	NAME OF COURSES	certification available?	Duration	Cost	Eligibility for certification	Remarks
5	CS201: Elementary Data Structures		38 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	Survey basic abstract data types, their associated algorithms, and how they are implemented. Topics discussed include the structures of stacks, queues, lists, sorting and selection, searching, graphs, and hashing; performance tradeoffs of different implementations; and asymptotic analysis of running time and memory usage.
6	Python for Data Science	Certificate Course	67 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn data science using the Python programming language by looking at data processing, data analysis, visualization, data mining, and statistical models. By the end of this course, you will be able to implement Python code for these data science topics.** This course attempts to strike a balance between presenting the vast set of methods within the field of data science and Python programming techniques for implementing them. Problem-solving and programming implementation will be emphasized throughout the course. All techniques presented will be introduced using real-world programming examples. A major goal of the course is to ensure that when you finish the course, you will have the programming and conceptual expertise you need to join the field of data science.
7	CS301: Computer Architecture	Certificate Course	48 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Explore hardware/software components, assembly language, and the functional architecture and design of computers, with a focus on topics like instruction sets, processor arithmetic and control, Von Neumann architecture, pipelining, memory management, storage, and input/output.** In this course, you will study the history of modern computing technology before learning about modern computer architecture and a number of its essential features, including instruction sets, processor arithmetic and control, the Von Neumann architecture, pipelining, memory management, storage, and other input/output topics. The course will conclude with a look at the recent switch from sequential processing to parallel processing by looking at the parallel computing models and their programming implications.
8	CS302: Software Engineering	Certificate Course	41 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn how to apply an engineering approach to computer software design and development by focusing on topics like life cycle models, software requirements, specification, conceptual model design, detailed design, validation and verification, design quality assurance, software design/development environments, and project management.** The course will begin with an introduction to software engineering, giving you a definition of this body of knowledge, as well as a discussion of the main methodologies of software engineering. You will then learn about the Software Development Life Cycle (SDLC), major methodologies followed by software modeling using Unified Modeling Language (UML), a standardized general-purpose modeling language used to create visual models of object-oriented software.
9	CS401: Operating Systems	Certificate Course	120 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Examine how operating systems and design have evolved as changes in hardware and software led to contemporary operating systems. Topics include basic OS concepts, methods of OS design and construction, process coordination, management, and algorithms for CPU scheduling, memory, and general resource allocation.** This course will introduce you to modern operating systems. We will focus on UNIX-based operating systems, though we will also learn about alternative operating systems, including Windows. The course will begin with an overview of the structure of modern operating systems. Over the course of the subsequent units, we will discuss the history of modern computers, analyze in detail each of the major components of an operating system (from processes to threads), and explore more advanced topics in the field, including memory management and file input/output. The class will conclude with a discussion of various system-related security issues.

Saylor Academy, USA (List of Certification Courses)

https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start

S.No.	NAME OF COURSES	certification available?	Duration	Cost	Eligibility for certification	Remarks
10	CS402: Computer Communications and Networks	Certificate Course	42 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Explore the hardware, software, and architectural components involved in computer communications in local area networks. Topics include the basics of computer networks, switching, routing, protocols, and security.** This course is designed to give you a clear understanding of how networks, from in-home local area networks, or LANs, to the massive and global Internet, are built and how they allow us to use computers to share information and communicate with one another.
11	CS403: Introduction to Modern Database Systems	Certificate Course	42 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn about database architecture and implementation by exploring Structured Query Language (SQL), including topics like file structures and access methods; database modeling, design, and user interface; the components of database management systems; and information storage and retrieval.** This course will provide a general overview of databases, introducing you to database history, modern database systems, the different models used to design a database, and Structured Query Language (SQL), which is the standard language used to access and manipulate databases. Many of the principles of database systems carry to other areas in computer science, especially operating systems. Databases are often thought of as one of the core computer science topics since many other areas in the discipline have been derived from this area.
12	CS406: Information Security	Certificate Course	46 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn the principles of information security to protect the confidentiality, integrity, and availability of information. Discuss the modes of threats and attacks on information systems, threat mitigation, cryptography, user identification and authentication, access control, privacy laws, and more.** This course covers information security principles, an area of study that engages in protecting the confidentiality, integrity, and availability of information. Information security continues to grow with advancements in technology – as technology advances, so do threats, attacks, and our efforts to mitigate them. In this course, we discuss the modes of threats and attacks on information systems. We also discuss an important area of threat mitigation that saw rapid development in the twentieth century: cryptography
13	PRDV410: Introduction to Java and Object-Oriented Programming	Certificate Course	6 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn the basics of object-oriented (OO) programming, a paradigm that uses objects that contain data in the form of instance variables and methods that control the behavior of the objects. This course is suitable for anyone and uses Java to demonstrate key OO concepts.** We will begin this course with an introduction to the difference between traditional structural program design and object-oriented (OO) programming. This course will explore the unique, dynamic approach that OO offers and discuss why it is an effective and flexible paradigm. We then move into the essential concepts that are part of all OO programming, including inheritance, polymorphism, abstract classes, encapsulation, and coupling.
14	PRDV401: Introduction to JavaScript I	Certificate Course	17 hours	Free	To receive a free Course Completion Certificate, you will need to earn a grade of 70% or higher on this final exam.Once you pass this final exam, you will be awarded a free Course Completion Certificate.	**Learn fundamental programming concepts using JavaScript to produce code for browser-based interactive experiences using conditions and events.** This course is the first in a series of three short courses that will build your programming skills in the JavaScript language. JavaScript is a client-side scripting language for producing interactive content in a web browser. It is one of the most popular programming languages worldwide. This course is for future JavaScript developers with some HTML and CSS knowledge. This course with a brief history of JavaScript. We then introduce programming concepts such as variables, control statements, objects, exception handling, and best practices. You will also learn about the Document Object Model or DOM, which provides the structure for dynamically displaying and manipulating elements on a webpage written in HTML using JavaScript.

Oracle -My Learn Courses

SNo	Topic	Duration	Module	Free/paid	Link
FOR CIVIL ENGINEERING STUDENTS					
1	Primavera P6 Professional	Number of Modules 4 Duration 02h 59m	Learn to plan, execute, and update a project. **Benefits to you:** Learn to plan, execute, and update a project. **Additional Information :** Learn about Oracle's Products : P6 Professional **Display By** : Free	Free	https://mylearn.oracle.com/ou/learning-path/primavera-p6-professional/94431
2	Migrating from P6 to Primavera Cloud	Number of Modules 1 Duration 19m	Videos compare features between P6 and Primavera Cloud to assist administrators whose organizations are migrating to Primavera Cloud. Benefits to you Videos compare features between P6 and Primavera Cloud to assist administrators whose organizations are migrating to Primavera Cloud.	Free	https://mylearn.oracle.com/ou/course/migrating-from-p6-to-primavera-cloud/112669
3	Primavera P6 Project Management	7 Modules 1h 3m	Learn to plan, execute, and update a project. Benefits to you **You will learn:** Getting Started, Planning the Project, Scheduling, Managing Resources, Executing the Project, Analyzing the Project, Additional Information, Learn about Oracle's Products : P6 **Display By** : Free	Free	https://mylearn.oracle.com/ou/course/primavera-p6-project-management/92446
4	Primavera P6 Application Administration	1 Module 14m	Benefits to you Primavera P6 Application Administration	Free	https://mylearn.oracle.com/ou/course/primavera-p6-application-administration/104213
5	Migrating from P6 to Primavera Cloud	1 Module 19m	Benefits to you Videos compare features between P6 and Primavera Cloud to assist administrators whose organizations are migrating to Primavera Cloud.	Free	https://mylearn.oracle.com/ou/course/migrating-from-p6-to-primavera-cloud/112669
6	Primavera P6 Visualizer	Number of Modules 1 Duration 51m	Benefits to you You will learn: **Overview and Navigation, Building a Gantt Chart to View Activities, Building a Gantt Chart to Measure Activity Progress, Building Timescaled Logic Diagram to Monitor a Portfolio, Building Timescaled Logic Diagram to Track Progress by Department, Building Timescaled Logic Diagram to Filter Activities, Additional Information, Learn about Oracle's Products : P6, P6 Professional** **Display By : Free**	Free	https://mylearn.oracle.com/ou/course/primavera-p6-visualizer/94433